INSTRUMENTATION COSTS OF METRICS

Concerning the costs of instrumenting code with metrics, we can distinguish two cases:

- Using auto-instrumentation usually requires little effort, such as including a library or configuring it (e.g., with Prometheus [http://mng.bz/7Dr9]; OpenTelemetry [https://opentelemetry.io/docs/instrumentation/]; and, increasingly, eBPF based).
- Manual instrumentation requires more effort than auto-instrumentation but also offers more control over which metrics you want to expose in which manner. Typically, these instrumentations should be one-offs, and for a class of metrics (business-level metrics), this kind of instrumentation is the only viable one.

The readability, and with it the maintainability, of the code can suffer with manual instrumentation.

Let's now have a concrete look at the instrumentation costs from a developer's point of view: the original code in listing 2.1 has 24 lines of code, and the code instrumented with metrics in listing 2.3 has 56 lines of code overall. To isolate the metrics-caused instrumentation costs from the total (logs + metrics) instrumentation cost, we need to subtract the 8 lines of code from the logs-caused instrumentation. We then arrive at a value of 48 lines of code.

Taking these two LOC numbers together, we can calculate the B2I ratio and arrive, for the metrics, at a value of 2. This B2I value is slightly higher than what we had with logs (1.33), but it is still not worrisome.

If you want to learn more about good practices concerning metrics instrumentation, I recommend you peruse the following resources:

- CNCF SIG Instrumentation guidance on instrumenting Kubernetes (http://mng.bz/mVer).
- CNCF TAG observability whitepaper input with guidelines for developers on how to implement new metrics (https://github.com/cncf/tag-observability/issues/18).

TELEMETRY COSTS OF METRICS

Agents for metrics have considerations similar to those that we discussed for logs. There is, in addition, the challenge around the cardinality—that is, the number of different values possible in the dimensions of a metric that can contribute to a (dramatic) increase of the costs.

Examples of metrics agents include the following:

- CNCF Prometheus, especially in agent mode (https://prometheus.io/blog/2021/11/16/agent/)
- The CNCF OpenTelemetry collector (https://opentelemetry.io/docs/collector/) and, more specifically, collectors from OpenTelemetry distributions, such as the AWS Distro for OpenTelemetry Collector (http://mng.bz/5wBB)